Daily Logic Drill #7

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Mixed Synthesis Capstone · Every technique from Days 1–6, combined.

Section A

1. What is the negation of the proposition "$\forall x \in \mathbb{R},\ x^2 + 1 > 0 \implies x > 0$"?

Ans: $\exists x \in \mathbb{R}$ such that $x^2 + 1 > 0$ and $x \leq 0$

2. For integers n, is "n is divisible by 6" necessary, sufficient, both, or neither for "n is divisible by 2"?

Ans: Sufficient only.

3. Consider the implication chain $P \implies Q$, $\neg R \implies \neg Q$, and $R \implies S$. If P is true, what is the truth value of S?

Ans: True.

4. A student solves $\sqrt{2x+3} = x$ by squaring both sides and obtains candidate solutions $x = 3$ and $x = -1$. Which candidate is extraneous?

Ans: -1

5. Find the smallest odd composite positive integer that is not a prime power.

Ans: 15

6. For real numbers x, is "$x^2 > 9$" necessary, sufficient, both, or neither for "$x > 3$"?

Ans: Necessary only.

7. A sequence satisfies $u_1 = 2$ and $u_{n+1} = 3u_n - 2$. What is the closed-form expression for u_n?

Ans: $3^{n-1} + 1$

8. In a proof by contradiction of the implication $P \implies (Q \vee R)$, what compound statement is assumed alongside P?

Ans: $\neg Q \wedge \neg R$ (both false).

9. Find the smallest positive integer n such that $2^n + 1$ is composite.

Ans: 3

10. If $P(1)$ is true and $P(k) \implies P(2k)$ holds for all $k \geq 1$, how many integers in $\{1, 2, \ldots, 20\}$ are proven true?

Ans: 5

Section B

1. Prove by contrapositive: "If n is not a multiple of 3 and not a multiple of 5, then n is not a multiple of 15."

Ans: Proof by contrapositive

2. Prove by contradiction: "There is no positive integer n such that $n^2 + n$ is odd."

Ans: Proof by contradiction

3. Prove by induction: $\sum_{i=1}^{n} i^2 = \dfrac{n(n+1)(2n+1)}{6}$ for all positive integers n.

Ans: Proof by induction

4. A student wants to prove "if n is not a multiple of 5, then n^2 is not a multiple of 25" by instead showing "if n is a multiple of 5, then n^2 is a multiple of 25." Is this a valid proof of the original claim? Justify your answer.

Ans: No.

5. Give a counterexample showing that "n is prime" is not sufficient for "n is odd."

Ans: $n = 2$

6. Negate the statement: "For every set A, there exists a set B such that $A \cap B = \emptyset$." Is the negated statement true or false?

Ans: Negation: "There exists a set A such that for every set B, $A \cap B \neq \emptyset$." This negation is false.

7. Prove by induction that $n^3 + 2n$ is divisible by 3 for all positive integers n.

Ans: Proof by induction

8. Prove by contradiction that $\sqrt{6}$ is irrational.

Ans: Proof by contradiction

9. For a polynomial f, classify "$f(x) > 0$ for all real x" as necessary, sufficient, both, or neither for "f has no real roots."

Ans: Sufficient but not necessary.

10. A student proves $P(n)$ for all $n \geq 1$ by checking $P(1)$ directly, then showing $P(k) \implies P(k+1)$ only for $k \geq 2$ (skipping $k = 1$). Explain the gap this leaves in the proof.

Ans: $P(2)$ is never derived, so the chain of implications never actually starts.

Section C

1. P: "n divisible by 4"; Q: "n divisible by 2 and by 6".

Ans: D

2. A student "proves" the converse of "every element even $\implies$ sum even" using only $S = \{2, 4\}$.

Ans: B

3. Which technique suits "there exists n with $n^2 + 3n + 1$ not prime"?

Ans: B

4. "Multiple of 9 $\iff$ digit sum multiple of 9", verified only for $n = 9, 18, 27$.

Ans: B

5. Contrapositive of "If n^2 is not divisible by 4, then n is odd".

Ans: B

6. Induction proof that $n^2 + 5n + 2$ is always even.

Ans: A

7. P: Euclid's Lemma (as a universal statement); Q: existence of a counterexample prime.

Ans: B

8. Necessity argument for "$x = 1 \iff x^3 - 1 = 0$".

Ans: B

Section D

1. Prove the treasure location (x, y) is uniquely determined by the pair $(x^2 + y,\ x + y^2)$, given the two values are distinct. *(after BMO1 2010 Q6)*

Ans: Proof below.

2. Bags with m, n balls; operations: remove equal amounts, or triple one bag. Prove $m - n$ odd $\implies$ can never reach $(0, 0)$. *(after BMO1 2012 Q4)*

Ans: Proof below, via the invariant $m - n \bmod 2$.

3. Prove no positive integers $a < b < c$, all even, satisfy $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$.

Ans: Proof below.

4. Prove the uniqueness part of the Fundamental Theorem of Arithmetic, given Euclid's Lemma.

Ans: Proof by strong induction (via minimal counterexample) below.

5. (a) Prove $mx + ny = 1 \implies \gcd(m, n) = 1$. (b) Deduce Euclid's Lemma from Bézout's Identity.

Ans: Proofs below.